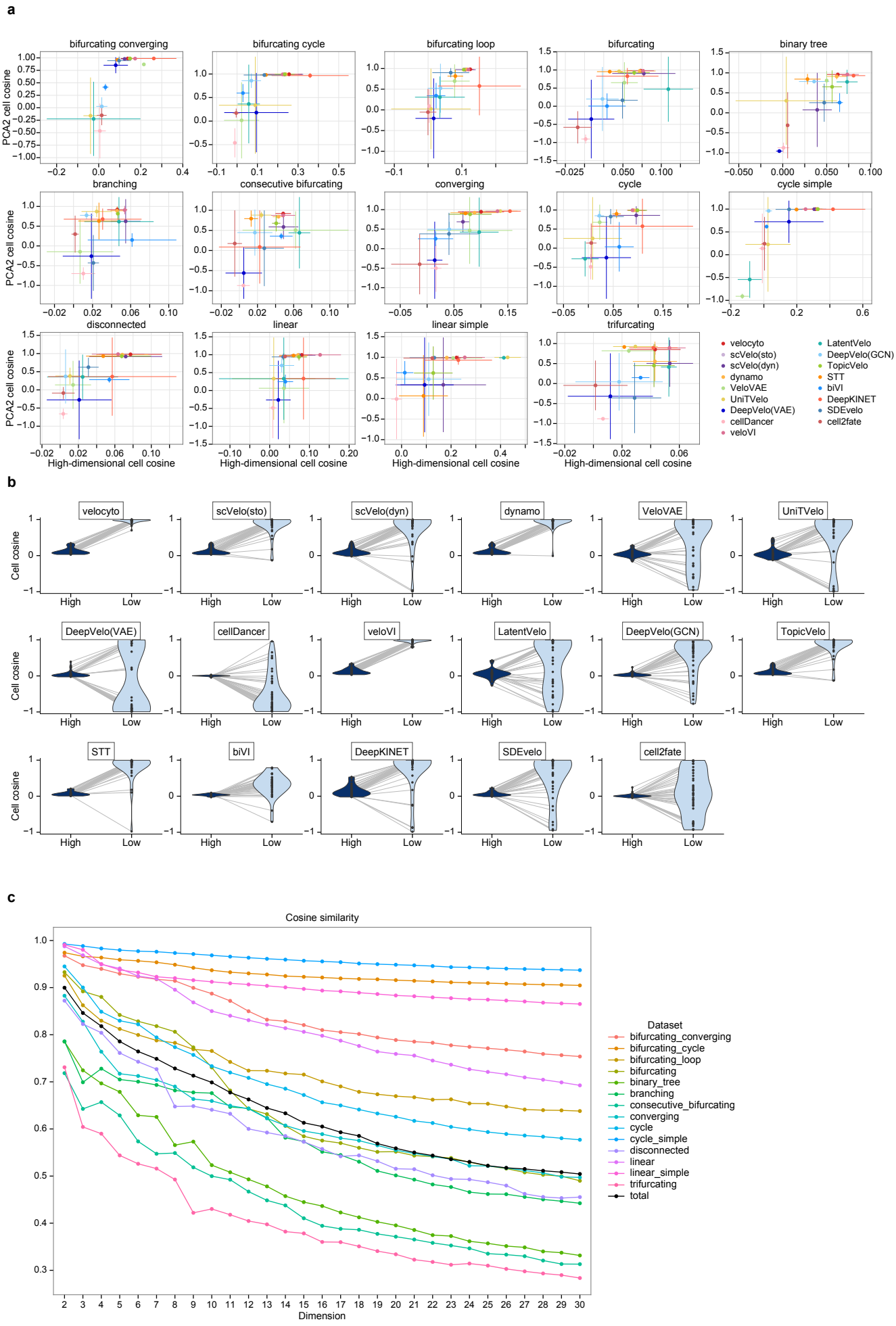

a
bifurcating converging
bifurcating cycle
bifurcating loop
bifurcating
binary tree
branching
consecutive bifurcating
converging
cycle
cycle simple
disconnected
linear
linear simple
trifurcating
PCA2 cell cosine
High-dimensional cell cosine
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
TopicVelo
STT
biVI
DeepKINET
SDEvelo
cell2fate
b
Cell cosine
High
Low
c
Cosine similarity
Dimension
Dataset
bifurcating_converging
bifurcating_cycle
bifurcating_loop
bifurcating
binary_tree
branching
consecutive_bifurcating
converging
cycle
cycle_simple
disconnected
linear
linear_simple
trifurcating
total

Extended Data Fig. 5 | High- and low-dimensional velocity evaluation

a, Original-style scatter plots separated by the 14 dyngen topologies. Each point summarizes one method across available matched seed datasets (mean and sample SD); high-dimensional and PCA2 scores are cell-level cosine similarities. b, Paired high/low-dimensional violin plots, faceted by method; lines join the same dataset, with dataset medians shown as points. c, Cosine similarity between predicted and simulated true RNA velocity in PCA spaces of 2–30 dimensions, restoring the original accuracy-curve definition. Each colored point is the median of finite method–dataset cell-level cosine scores within one topology at that dimension; the black total curve is the median across all 42 datasets and available methods. These are projected velocity-accuracy scores, distinct from the high/low-dimensional score-vector consistency in Fig. 3d. Seventeen available configurations are included; missing scores are excluded without imputation. Source scores and valid counts are supplied. TopicVelo table values retain unverified native-transition provenance. No model was rerun.